2.1 · Scaling Laws and the True Cost of Data

Week 2 · Tuesday Jul 21 — Data, Environment, and Human-Centered Sustainable AI

Generative AI & Sustainability · Munich Summer School 2026

Outline

- Scaling laws, briefly (Week 1 recap): predictable improvement, the Chinchilla balance, and where the returns flatten
- Why frontier models are data-hungry, and quality versus quantity of training data
- Where training data comes from: web scrapes, licensed corpora, user data, and human annotation and RLHF labor
- Sustainability framing: the compute and human cost of scaling, and why bigger is not automatically better
- Bridge to the 10:10 discussion: what published "training cost" figures leave out — development runs, hardware manufacturing, and water — as measured in last night's reading

Yesterday's open question

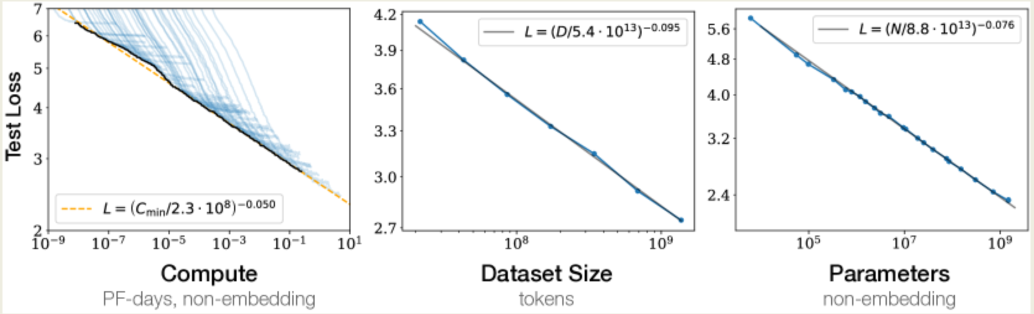
- Monday: an LLM is a next-token predictor whose behavior comes from **pre-training on enormous text**
- Today's questions:
  - why *enormous*? Who decided bigger is better?
  - where does all that text **come from**?
  - who **pays** — in compute, water, and working hours?
- Tonight's admission tickets stay on your desk — we use them at 10:10

Scaling laws

A 12-minute recap — you've seen these curves twice already. Today: what they imply.

The finding, in one slide (recap)

- You saw these charts last week: prediction error falls **smoothly and predictably** with model size, data, and compute — a straight line on log axes across **seven orders of magnitude**
- Why it changed everything: better AI stopped needing new ideas and became a **purchasing problem** — same architecture, 10× compute, better model
- Labs could **forecast** quality before spending the money — so they spent the money



- **A scientific curve became a business plan — and an energy-demand curve.**

Balance matters: parameters *and* tokens

- 2022 "Chinchilla" result: many models were **too big and under-fed**
- For a fixed compute budget, best results come from balancing model size with training data — roughly **20 tokens per parameter**
- A 70B-parameter model trained on *more data* beat a 280B-parameter model
- Consequence: the data appetite **exploded**
  - recent open models train on **~15 trillion tokens**

Where returns flatten

- The law is a **power law**: each equal quality step costs **~10× more** compute
- Log axes flatter progress — linear axes show a wall of diminishing returns
- Meanwhile costs grow *linearly with every token processed*: energy, water, hardware, money
- Design question for your startups: is the last 2% of quality worth 10× the footprint — **for a campus learning space?**

The data behind the models

~15 trillion tokens. Somebody wrote every one of them.

How much text is 15 trillion tokens?

- All of **English Wikipedia**: ~4–5 billion words — well **under 1%** of a frontier training corpus
- The rest: **web scrapes** at internet scale — filtered from Common Crawl (petabytes crawled monthly), plus code, books, papers, forums
- No human could ever read the corpus — **nobody fully knows what's in it**
- That gap — "we trained on it but can't inventory it" — is tomorrow's leakage lecture and today's 11:20 topic

Where training data comes from

Source	Examples	Open questions
Web scrapes	Common Crawl, forums, blogs	consent? copyright? quality?
Licensed corpora	news archives, stock photos, publishers	who gets paid — author or platform?
User data	chat logs, feedback, uploaded docs	did users understand the trade?
Synthetic data	model-generated text	quality drift, feedback loops
Human annotation	preference rankings, safety labels	whose judgment? whose working conditions?

Quality vs. quantity

- Raw web text is mostly **unusable**: spam, duplicates, boilerplate, toxicity
- The real craft is **curation**: filtering, deduplication, mixing sources — labs guard these recipes more closely than architectures
- Better data beats more data: small models on curated corpora rival much larger ones
- And the well is finite: projections say **high-quality public text runs out within this decade** — hence licensing deals, user data, and synthetic text

The human cost

"Human feedback" is not a metaphor. It is a workforce.

The labor behind the labels

- Step 3 of Monday's pipeline — human feedback — means **people, at scale**: ranking outputs, writing demonstrations, labeling harmful content
- Much of it outsourced to low-wage annotation workforces — investigations found content-safety labelers in Kenya paid **under $2/hour** to read the internet's worst material
- The polish of a frontier chatbot **embodies** those working hours — uncounted in any FLOP or kWh figure

Consent and compensation

- The training set almost certainly includes text by **you** — a forum post, a thesis, a review
- **Public ≠ permitted:** visible on the web is not the same as licensed for training (lawsuits from news publishers, authors, and artists are testing exactly this line)
- "Sustainable AI" in this course includes the **human supply chain**: consent, credit, compensation, working conditions
- You met this as **data provenance & consent** in Week 1's Ethical Design Checklist

What it costs the planet

The bridge to your reading — and the 10:10 discussion.

What the published numbers show — and hide

- When training costs are disclosed at all, it's usually **one number: the final training run**
- Last night's paper measured what that number leaves out:
  - **development** — the failed and experimental runs before the final one
  - **hardware manufacturing** — embodied carbon of the GPUs
  - **water** — cooling the data center and the power plant behind it
- Keep your headline number from the admission ticket in mind

Honest accounting strengthens your pitch

- Your university client will ask you to **account for your system's footprint** — as you began doing in week 1
- Today's lesson extends yesterday's:
  - scale has a **price curve** — locate your proposal on it
  - data has **owners and workers** behind it
  - published footprints are usually **undercounts** — ask what's missing
- Honest accounting can **differentiate Thursday's pitch** — few AI products offer it

Wrap-up

- Scaling laws made model quality purchasable — and made energy, data, and labor the real inputs of AI
- Frontier corpora are trillion-token web-scale collections nobody can fully audit, refined by **human** labor that rarely appears in the accounting
- "Bigger is better" flattens into "bigger is *marginally* better, much more expensive" — right-sizing is the sustainable design move
- **After the break:** admission tickets out — we discuss what a *complete* environmental accounting of an LLM looks like (Morrison et al.)